

King Khalid University
College of Computer Science
Department of Computer Science



جامعة الملك خالد
كلية علوم الحاسب
قسم علوم الحاسب

STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. Aymen TRIGUI

Course Code: 462 CCS-3

Student ID: 445801958

Section Number: 903

S. No

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LAB ACTIVITY - 6

(3 Marks)

Write a Python program to demonstrate a comparative analysis between the KNN algorithm and Logistic Regression. This activity should focus on cross-validation and associate tools to select between the two models.

Note that this dataset can also be loaded from the scikit_learn library using the following Python instructions:

```
from sklearn.datasets import load_iris
```

Activity Requirement:

- 1) first load the Iris dataset and perform feature scaling using StandardScaler. Then, split the dataset into training and test sets using train_test_split.
- 2) Create the KNN classifier with KNeighborsClassifier and the Logistic Regression classifier with LogisticRegression.
- 3) Use cross_val_score to perform 5-fold cross-validation on both models and obtain the cross-validation scores.
- 4) Finally, compare the mean cross-validation scores of the two models and select the model with the higher score as the better-performing model.

5-fold cross-validation results:

- KNN mean score: 0.9667

- Logistic regression mean score: 0.9533

Selected model: KNN